length field is up to the discretion of the sender, implying that a sender can choose to send more length octets than strictly necessary to encode the value.

A.10. End of Container Encoding

The end of a container type is marked with a special element called the end-of-container element. The end-of-container element is encoded as a single control octet with the value 18h. The tag control bits within the control octet SHALL be set to zero, implying that end-of-container element can never have a tag.

Control Octet
1 octet

A.11. Value Encodings

A.11.1. Integers

An integer element is encoded as follows:

Control Octet	Tag	Value
1 octet	0 to 8 octets	1, 2, 4 or 8 octets

The number of octets in the value field is indicated by the element type field within the control octet. The choice of value octet count is at the sender's discretion, implying that a sender is free to send more octets than strictly necessary to encode the value. Within the value octets, the integer value is encoded in little-endian format (two's complement format for signed integers).

A.11.2. UTF-8 and Octet Strings

UTF-8 and octet strings are encoded as follows:

Control Octet	Tag	Length	Value
1 octet	0 to 8 octets	1 to 8 octets	0 to 2^{64}-1 octets

The length field of a UTF-8 or octet string encodes the number of octets (not characters) present in the value field. The number of octets in the length field is implied by the type specified in the element type field (within the control octet).

For octet strings, the value can be any arbitrary sequence of octets. For UTF-8 strings, the value octets SHALL encode a valid UTF-8 character (code points) sequence. Senders SHALL NOT include a terminating null character to mark the end of a string.

A.11.3. Booleans

Boolean elements are encoded as follows:

